Adithya Rao Kalathur

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LinkedIn | GitHub | Portfolio

Profile

M.E. Computer Science student with experience in **machine learning, backend development, and systems programming**. Proficient in **Linux, Python, and Docker-based deployment**, with hands-on experience building scalable applications and ML pipelines. Strong focus on **debugging, performance optimization, and real-world system integration**. Seeking roles in **ML, backend, or systems engineering**.

Education

Manipal School of Information Sciences (MSIS) — Manipal, India
Master of Engineering in Computer Science and Engineering; CGPA: 7.96/10 — *2025 – 2027 (Expected)*

Manipal Institute of Technology (MIT) — Manipal, India
Bachelor of Technology in Computer and Communication Engineering; CGPA: 7.06/10 — *2020 – 2024*

Technical Skills

Languages: Python, SQL, Java, C
Machine Learning & Deep Learning: PyTorch (proficient), TensorFlow (familiar), Scikit-Learn, Neural Networks, CNNs, U-Net, Transfer Learning, Grad-CAM, Hugging Face, BiLSTM, XGBoost
NLP & Data Science: Natural Language Processing, Text Classification, Pandas, NumPy, Data Cleaning, Data Visualization
Data & Analytics: Power BI, DAX, ETL Pipelines, SQL
Backend & Web: React.js, Node.js, PostgreSQL, REST APIs, HTML/CSS
Systems & Tools: Linux, Shell Scripting, Git/GitHub, Docker, Debugging, API Integration, OpenCV

Work Experience

Full Stack Web Development Intern — Jun 2023 – Jul 2023
Thaniya Technologies [Certificate] — *Mangalore, India*

- Built and deployed a full-stack movie review and recommendation platform using **React.js, Node.js, and PostgreSQL**; shipped to production on Render via cloud hosting [GitHub].
- Designed the relational database schema, implemented REST APIs for all CRUD operations, and integrated backend services with the React frontend.
- Built content-based recommendation logic and owned the full delivery cycle — UI design to database integration to deployment.

Freelance Web Developer — Dec 2024 – Mar 2025
Startup Client Project — *Remote*

- Delivered a client-facing portfolio website for a landscape design startup end-to-end — requirements gathering, content architecture, responsive layout design, and deployment — within a fixed project timeline.
- Managed all stakeholder communication, iterated on designs based on client feedback, and structured service and contact pages for desktop and mobile.

Projects

Deep Learning-Based SCC Grading from Histopathology Images [GitHub] — 2025
Deep Learning, Transfer Learning, Computer Vision, Explainable AI (XAI), Grad-CAM, Gradio, PyTorch

- Benchmarked 5 transfer-learning CNN backbones (InceptionV3, EfficientNetB0, DenseNet121, MobileNetV2, ConvNeXt-Tiny); best model achieved **96.5% test accuracy** and **macro-F1 0.94** on a held-out dataset.
- Deployed an interactive Gradio web app with real-time Grad-CAM visualizations and LLM-generated prediction summaries for interpretability.
- Designed full ML pipeline: dataset splitting (70/15/15), class-weighted loss, and extensive augmentation to address class imbalance.

- Applied early stopping, learning rate scheduling and model checkpointing to improve generalization and reduce overfitting.

Night Vision Enhancement using Deep Learning [GitHub] 2025

U-Net, ResNet34, PyTorch, OpenCV, Computer Vision, Image-to-Image Translation

- Built two low-light image enhancement pipelines — a vanilla U-Net and a ResNet34 encoder-decoder U-Net — trained on the LOL dataset (485 pairs) and the ImageNet Night Vision dataset (40K+ pairs) using paired image-to-image translation.
- Achieved **PSNR of 18.84 dB** and **SSIM of 0.71** on the ImageNet Night Vision test set (5000 images) with the pretrained ResNet34 U-Net trained using L1 loss and ReduceLROnPlateau scheduling.
- Designed training pipelines with combined MSE + SSIM loss, ColorJitter augmentation, and residual learning; evaluated outputs quantitatively using PSNR and SSIM across multiple model variants.

Network Intrusion Detection System [GitHub] 2025

Random Forest, Gradient Boosting, Scikit-Learn, Feature Engineering

- Built an ML-based intrusion detection pipeline using Random Forest and Gradient Boosting, achieving **94%+ accuracy** with **<5% false positive rate** on network traffic classification.
- Performed feature engineering on traffic attributes including packet size, flow duration, and protocol flags; used feature importance analysis to surface the most discriminative classification signals.
- Evaluated model performance using confusion matrix, precision, recall, F1-score, and ROC-AUC on a held-out test set.

FMCG Supply Chain Analytics Dashboard [GitHub] 2025

Power BI, DAX, SQL, ETL, Data Analytics

- Built an end-to-end Power BI dashboard tracking supply chain KPIs — OTIF, LIFR, and VOFR — across 3 cities and 50+ product categories for Atliq Mart.
- Developed 20+ DAX measures and SQL-based ETL pipelines for root-cause analysis; identified a **15% fulfillment service gap** across key accounts that directly informed operational strategy.

Publications & Research

SQL Injection Detection: Comparative Study of SVM, RF and Hybrid Models [Publication] Oct 2025

SVM, Random Forest, BiLSTM, NLP, Hybrid Ensemble, Cybersecurity

- **First Author** — Published as a book chapter in *Connecting Intelligence: Trends in Computation and Data Communication*, Taylor & Francis / CRC Press (ISBN: 9781003773504); presented at ICCIDC 2025, Bali, Indonesia [Certificate]
- Designed a hybrid SVM-RF ensemble for automated SQL injection detection on 30,920 labeled queries, achieving **97.84% accuracy**, **98.09% precision**, and **97.57% recall**. Evaluated a BiLSTM model reaching **98.75% accuracy**, demonstrating the effectiveness of sequence-based NLP for threat detection.

Certifications

- IBM Data Science Professional Certificate — Coursera, 2024
- Google Data Analytics Professional Certificate — Coursera, 2024
- Generative AI with Large Language Models — DeepLearning.AI, 2025
- Google Cybersecurity Professional Certificate — Coursera, 2024

View All Certificates

Languages

English: Professional proficiency
Kannada: Native
Tulu: Native
Hindi: Conversational